Trafficelligence - Project Report

1. INTRODUCTION

1.1 Project Overview

Trafficelligence is an AI-powered traffic volume prediction system designed to optimize urban traffic flow. It utilizes machine learning models to forecast vehicle volume based on historical and real-time traffic data, helping in traffic management and city planning.

1.2 Purpose

The purpose of this project is to develop an intelligent system that can predict traffic volumes at different times and locations, thereby aiding commuters, traffic authorities, and smart city initiatives in efficient traffic handling.

2. IDEATION PHASE

2.1 Problem Statement

Urban traffic congestion is a major issue causing delays, fuel wastage, and environmental pollution. Predicting traffic volume in advance can enable better traffic routing and planning.

2.2 Empathy Map Canvas

Empathy map focused on understanding city commuters' experiences, challenges (e.g., traffic delays, route uncertainty), and needs (e.g., accurate traffic predictions).

2.3 Brainstorming

We brainstormed various solutions including traffic cameras, sensors, and ML models. We chose a data-driven prediction model using regression algorithms due to its scalability and cost-efficiency.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey begins with a user launching the app, entering destination info, and receiving traffic volume predictions. The system processes location data and traffic datasets using ML models to generate output.

3.2 Solution Requirement

Functional: Registration, login, data input, prediction output.
Non-Functional: Usability, performance, security, scalability.

3.3 Data Flow Diagram

The DFD includes user input, processing by ML model, data retrieval from dataset, and result output to user.

3.4 Technology Stack

Frontend: HTML/CSS/JavaScript
Backend: Python (Flask)
ML Libraries: Scikit-learn, XGBoost
Database: CSV Dataset
Deployment: Localhost / Cloud (optional)

4. PROJECT DESIGN

4.1 Problem Solution Fit

The ML approach provides a cost-effective and scalable way to forecast traffic, addressing real user pain points.

4.2 Proposed Solution

Develop a predictive system using regression models such as Linear Regression, Decision Tree, Random Forest, SVR, and XGBoost to estimate traffic volume.

4.3 Solution Architecture

Client Input -> Backend Server -> Model Inference -> Output Response
Models and dataset are processed on the server to predict traffic volume.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Week 1: Problem identification & data collection
Week 2: Data cleaning and preprocessing
Week 3: Model development
Week 4: Evaluation and result analysis
Week 5: UI integration and final testing

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Tested ML models using metrics like MAE, RMSE, and R^2 score to compare prediction accuracy. Random Forest and XGBoost showed highest performance.

7. RESULTS

7.1 Output Screenshots

Included screenshots of model predictions, graphs, and dashboards in the final report (attach in Appendix if applicable).

8. ADVANTAGES & DISADVANTAGES

Advantages:
- Helps in better traffic management
- Scalable solution for smart cities
- Cost-effective using existing data

Disadvantages:
- Prediction accuracy depends on data quality
- Doesn't account for accidents or weather unless integrated

9. CONCLUSION

Trafficelligence provides a smart, data-driven approach to traffic prediction. With continued data integration and expansion, it can support urban traffic systems efficiently.

10. FUTURE SCOPE

Incorporate real-time GPS data, weather impact analysis, and integration with smart traffic lights to enhance prediction and response capabilities.

11. APPENDIX

Source Code: Attached separately
Dataset Link: Available on UCI repository (Metro Interstate Traffic Volume)
GitHub & Project Demo Link: [To be updated by user]